Cross-Atlas ASD Prediction from Resting-State fMRI: Machine Learning Stability and Neural Network Sensitivity

Abstract

Background: Autism Spectrum Disorder (ASD) is characterized by difficulties in social interaction and restricted or repetitive behaviors. Early and accurate identification improves outcomes, and neuroimaging-based machine learning approaches have shown promise in supporting diagnosis. Current pipelines, however, rely on either/or selections of feature extraction and classification methods, limiting cross-comparative evaluation.

Objectives: This study aimed to evaluate ASD classification performance across multiple machine learning models and a basic digital neural network (A1DNN) using two brain atlases from the ABIDE resting-state fMRI dataset, and to determine whether atlas selection differentially affects model performance.

Methods: Resting-state fMRI data from the ABIDE repository were processed using two brain atlases: Eickhoff-Zilles (EZ) and Craddock 200 (CC200). Principal Component Analysis (PCA) retaining 95% of explained variance was followed by Recursive Feature Elimination with Cross Validation (RFECV) using an AdaBoost meta-classifier. A battery of machine learning classifiers was evaluated, including KNN, Logistic Regression, Decision Tree, Random Forest, three Support Vector Classification variants, Stochastic Gradient Descent, Gaussian Naïve Bayes, Voting Classifier,

and Stacking Classifier. An A1 Digital Neural Network (A1DNN) was additionally implemented using TensorFlow and Keras for comparison. All models used stratified 10-fold cross validation.

Results: Machine learning model performance was equivalent across both atlases. Liblinear SVC achieved the highest test accuracy (72%) with 100% test recall. The A1DNN showed atlas sensitivity: CC200 outperformed EZ on test accuracy (63% vs. 53%) and precision (61% vs. 49%), while recall was equivalent (54%) across both.

Discussion: The equivalence of machine learning performance across atlases suggests that the PCA-RFECV pipeline normalizes atlas-level differences prior to classification, while the A1DNN's sensitivity to atlas selection indicates that neural network architectures retain meaningful responses to parcellation choice. These findings have practical implications for pipeline design in neuroimaging-based ASD classification.

Conclusions: Atlas selection affects neural network but not machine learning classifier performance for ASD prediction from resting-state fMRI. The PCA-RFECV pipeline provides a computationally efficient alternative to SVM-based feature selection, enabling broader cross-comparative evaluation. Future work will extend atlas comparisons across all ABIDE atlases using the same pipeline.

Keywords: Autism Spectrum Disorder; resting-state fMRI; machine learning; deep learning; ABIDE; Support Vector Classification; Principal Component Analysis; neural network.

1. INTRODUCTION

Autism Spectrum Disorder (ASD) is denoted as a significant impairment in social, occupational, or other areas of day-to-day function [1, 2]. Neurobiological differences that meet diagnostic criteria for ASD are found within social difficulties — often types of disconnect between the individual and their environment — and repetitive behaviors, including strong orientation toward sameness, limited interests, or physical and mental repetitions [1]. Clinical screenings face the difficulties of exclusive use of self-report measures not being viable, and conceptualization of behavioral characterizations changing over time [3]. Additionally, quick evaluations can be high in both false positives and false negatives, via false attribution to common considerations, such as a task that is too developmentally advanced for a person of a specific age [3].

While traditionally autism has been diagnosed via psychological screenings, its neurobiological basis has led to the use of machine learning and deep learning models with resting-state fMRI images to both substantiate the connection between the two and to provide accurate assessment through direct medical and empirical data [4]. ASD classification from neuroimaging data is inherently a pipeline problem: information preprocessing, feature selection, classification, and evaluation each carry trade-offs in terms of computational cost and model accuracy [4]. Standard approaches such as Support Vector Machine (SVM) are the most common, with gradient and ensemble tree-based classifiers a close second, meaning the strengths of one approach are not available when the other is selected in isolation. The same holds for neural network approaches, which represent variations on a short list of architectures [4].

The present work addresses this limitation by optimizing computational budget to enable cross-comparison across machine learning and deep learning model types against two commonly used atlases for ABIDE resting-state fMRI data: the Eickhoff-Zilles (EZ) and Craddock 200 (CC200) atlases. Feature selection uses Principal Component Analysis (PCA) fol-

lowed by Recursive Feature Elimination with Cross Validation (RFECV) as an alternative to SVM-based feature selection, addressing known limitations of the latter including computational cost and class-label-blind dimensionality reduction [5, 6]. The EZ atlas parcellates the brain into cytoarchitectonically defined regions derived from histological mapping, whereas the CC200 atlas uses a data-driven clustering approach to define 200 functionally homogeneous regions of interest. These two atlases therefore represent structurally and methodologically distinct approaches to brain parcellation, making their cross-comparative evaluation a meaningful test of atlas sensitivity in ASD classification pipelines. The significance of this work lies in demonstrating how updating fundamental steps in the machine learning pipeline — particularly feature selection — can provide more thorough, cross-comparative evaluation of both machine learning and deep learning models for ASD prediction from resting-state fMRI data.

2. LITERATURE REVIEW

Autism Spectrum Disorder is characterized by significant difficulties in social interactions and limited or repetitive behaviors that cause difficulties in relating and functioning within environments [1]. The amount of assistance individuals may require varies widely; many individuals become functional community members but face ongoing challenges with employment and independent living. Diagnostic practices based on age and developmental group also vary, given that individuals may or may not receive screening depending on circumstance and individual characteristics [1]. This variation underlies disagreements such as whether autism should be considered a condition or disorder, and whether neurobiological aptitudes such as attention to detail should be classified under the same nomenclature or separately [1].

Given both the difficulties in clinical diagnosis and innovations in non-invasive brain imaging techniques, regions of interest within fMRI images can be classified via machine learning and artificial intelligence algorithms [7]. The common methodologies in prediction

of autism via fMRI images [4] provide insight into what is considered standard practice and where further improvement is needed. It should be noted that current work has a male and higher-functioning bias in data due to data collection and compliance constraints, which is compounded by the use of this data to build models where evaluations with more representative data are not yet viable. Two main approaches to rs-fMRI classification exist: supervised learning, which uses functional connectivity and statistical spatial relationships, and unsupervised learning, which is oriented around functional organization of brain dynamics [8]. Common methodologies follow standard practices for binary classification, with SVM being the most common for machine learning models and CNN being the most common for neural networks [9, 10]. Future work in this area lies not only in corrections of the biases indicated above, but also in the development and implementation of techniques for proper identification of subtle and complex patterns in data [4]. Feature selection and retention must be optimized for discriminant features while minimizing inaccuracy, redundancy, and computational cost. Classification modeling operates on discrete rather than continuous prediction — in this case binary ASD/Non-ASD classification [5]. While use of auto-encoders that identify and use latent traits for classification has shown promise, this approach has not yet been demonstrated in larger multi-site data [7].

2.1. Machine Learning Models

Resting-state fMRI (rs-fMRI) relies on spatial relationship mapping from blood flow in regions of the brain in real time, with spatially distinct but functionally related areas determined by seed-based analysis. As a promising alternative to seed-based analysis, Independent Component Analysis arose in the early 2000s, with other machine learning models continuing to develop throughout the decade [6]. Unsupervised machine learning models (clustering, latent, and non-linear embedding) function in spatial relationships, covering both spatially distributed functional relationships and parcellation of brain regions. Atlases have not demonstrably been shown to have a single best option — they represent organizations of

brain mapping relevant to a given study — but clustering adapts to define regions without generalizing well to test data, while decomposition does the opposite, creating a trade-off for model selection [6]. Features must be drawn from images with uniformity and consistency as key considerations to ensure validity of training and test data [5]. The most common means of reducing features is dimensionality reduction, which produces the same number of rows but a reduced number of columns. While this is a very common technique [5], improvements are not guaranteed since class labels are ignored. Results are commonly evaluated by metrics such as accuracy, true positives and true negatives, and ROC curves. Gaussian Naïve Bayes (GNB) is useful for highly iterative classification, but both GNB and Fisher's Linear Discriminant Analysis (LDA) face issues with noise in feature selection [5]. Logistic Regression and Support Vector Classification techniques outperform GNB and LDA, but while the former lends itself to multi-class settings, common SVC packages require binary classification to be produced in multiple one-versus-one sets [5].

Spatially related features in resting-state fMRI are in part possible because each person has a distinct but consistent connectome. Unsupervised learning commonly performs recognition-based classification via clustering (e.g., Nearest Neighbors) and decomposition (e.g., Principal Component Analysis) [6]. Supervised learning commonly uses atlases for extracting supervoxel information with pipelines that delineate connectivity, with correlation matrices as the most common type of functional connectivity matrix for train/test and prediction. Feature selection is typically drawn from sparse spatial relationships (LASSO), dimensionality reduction (PCA), and/or implicit regularization (SVMs). Techniques such as Logistic Regression, DNN, SVM, and tree-based classifiers are used for prediction [6, 11]. Rimal et al. [12] demonstrated the use of Logistic Classification, Linear and Kernel Support Vector Classification, and Random Forest Classification for ASD prediction. As a conceptual continuation of this work, SVM for rank-order feature selection followed by RFE for feature reduction were used by Mohan and Paramasivam [10]. Binary classification with within/between relationships was examined by SVM, IBk, and Naïve Bayes with standard

accuracy metrics [2].

2.2. Deep Learning Models

While machine learning techniques rely on explicit feature extraction and selection, deep learning performs these steps in an integrated manner [13]. The complexity of neuroimaging data, however, can make deep learning optimization difficult [13]. Common trends in deep learning for predicting autism indicate CNN and RNN variants as holding the most promise in terms of consistent performance, whereas variations such as GANs produce results that can vary widely [14, 15, 16]. Model parameters and network structure parameters are the paramount considerations for current and future improvements [14]. Wang et al. [9] used SVM-RFE for feature selection with LSTM and GRU for neural networks, with correlation matrices drawn from the CC200 atlas from ABIDE. Ahmed et al. [17] explored multiple preprocessing pipelines, including 3D CNN and RNN with LSTM. The Configurable Pipeline for the Analysis of Connectomes (CPAC) preprocessing pipeline via TensorFlow has shown the most viable performance metrics for ABIDE rs-fMRI ROIs. In cross-comparison of atlases, Craddock 200 and Craddock 400 proved to have the highest performance metrics [11].

2.3. Problem Statement

The issue within the sources above is a lack of hybridization of techniques that substantially addresses both classification performance and statistical inference. This is most easily represented by the either/or selection between ICA and PCA [6]. Many of the above designs rely strictly on a train/test split, which can produce a see-saw effect on bias/variance; using a train/validation/test split instead allows for further optimization of these trade-offs [9]. In terms of machine learning, the sources above repeatedly demonstrate that Lasso, SVC, and tree-based models perform best for ASD/rs-fMRI prediction, but cross-comparative evaluation has not been performed with computational optimization. Likewise, for deep learning,

GRU was used alongside LSTM in some work, while other models were run on basic DNN with dropout. Cross-comparative evaluation of both PCA and ICA via DNN by LSTM is not present in the above work [12].

2.4. Research Questions

Areas of interest for this research are: (a) what improvements can be made in terms of computational budget optimization to allow for more thorough exploration of ASD prediction from the EZ and CC200 atlases; (b) what predictions and inferences can be made within and between machine learning models and the A1DNN; and (c) what differences in performance metrics can be observed between atlas selections for machine learning models versus the A1DNN?

3. METHODS

For preparation for model prediction, the CC200 and EZ atlases were used, with a Principal Component Analysis (PCA) selection criterion of explained variance of 95% or above applied to both [6]. The ABIDE dataset consists of resting-state fMRI data contributed by multiple international imaging sites, with participants including both ASD-diagnosed individuals and neurotypical controls. Preprocessed functional connectivity matrices for both atlas parcellations were obtained directly from the ABIDE repository, providing region-of-interest time series data that served as the input feature space for all models. PCA minimizes the reconstruction error of the data by using a linear projection for dimension reduction. This was then fed into Recursive Feature Elimination with Cross Validation (RFECV) using an AdaBoost meta-classifier. AdaBoost is a meta-classifier that applies gradient boosting using copies of a defined estimator along with a weight-adjusted learning algorithm in continuous iterations until no further adjustment is possible [5]. Stratified K-fold cross validation (leaving out 10% of examples per fold) was used for the RFECV and all subsequent cross validation within the models. Each machine learning model used an initial fitting and predic-

tion, followed by grid search with stratified K-fold ($k = 10$) to determine optimal parameters for train and test predictions. The predictive models selected include [18]: K-Nearest Neighbors (KNN), Multiclass Logistic Regression, Decision Tree, Random Forest, three Support Vector Classification variants (Linear, Liblinear, and Gaussian RBF), Averaged Stochastic Gradient Descent, and Gaussian Naïve Bayes. These were further used as base estimators for a Voting Classifier (hard voting) and a Stacking Classifier (final estimator: Multiclass Logistic Regression). Each model followed a train-fit, train-predict, and test-predict sequence, with AUROC curve generation and score output. A summary table of performance metrics was generated for each atlas.

3.1. Machine Learning Models

K-Nearest Neighbors classifiers take k as a user-defined constant and predict from each new reference point [18]. Since the model is non-parametric, training data is retained in full. Logistic Regression operates on binary classification where Y belongs to a particular category [18]. Support Vector Classifiers operate by maximizing the separation margin between the classification boundary and the nearest training data points [8]. Decision Tree Classifier uses piecewise comparisons with simple decision rules that increase in complexity according to tree depth [18]. Random Forest combines multiple decision trees and takes a majority vote across B trees to produce a more accurate and stable prediction; either full or sub-sampled bootstraps may be used [18]. Support Vector Classifier variants using the linear kernel apply a squared hinge loss method with regularization weights; Liblinear is optimized specifically for linear classification [18]. Stochastic Gradient Descent uses a one-versus-all classification scheme in which coefficients from one loss step are set to the average coefficients across all updates [18]. Naïve Bayes classifiers apply the assumption of conditional independence between every pair of features given the class label; Gaussian Naïve Bayes uses a Gaussian distribution over continuous features [18]. Voting Classifier uses ensemble combination by performing a hard vote across the preceding classifiers to minimize individual weaknesses

[18]. Stacking Classifier combines the preceding estimators to reduce inter-estimator bias, producing a final estimator trained through cross validation [18].

3.2. Deep Learning Model

The A1 Digital Neural Network (A1DNN) was created from the original feature and target datasets using TensorFlow and Keras as a baseline comparison against the machine learning classifiers above. These datasets followed analogous PCA procedures and subsequent train/test split parameters, along with metric selections from Keras (Accuracy, True Positives/Negatives, False Positives/Negatives, Precision, Recall, and AUC). A Sequential model was selected: Dense(64, input_shape = n_features), followed by Dense(32), Dense(16), Dense(8) layers with ReLU activation, and a final Dense(1) with sigmoid activation. The model was compiled using binary cross-entropy loss, the Adam optimizer, and the pre-defined metrics. Model fit used 50 epochs, batch size of 16, and a held-out validation set (x_{val}, y_{val}). Predictions were performed on the top five train and test cases (threshold > 0.05), classification accuracy for ASD/non-ASD was generated, evaluation was performed on test data, and cross validation using the same stratified K-fold procedure was performed on the full and training datasets respectively. Outputs for Accuracy, TP, TN, FP, FN, Sensitivity, and Specificity were generated for each atlas.

4. RESULTS

For all machine learning models using both EZ and CC200 atlases, performance metrics were equivalent across atlas selections. AUROC scores matched accuracy percentages for all models. Liblinear SVC achieved the best performance metrics overall. Multiclass Logistic Regression and Linear SVC were tied for second, followed by remaining SVC variants. The Voting Classifier and Stacking Classifier, which incorporate the preceding classifiers as ensemble components, also performed strongly. Each of these top-performing models achieved a test recall of 100% (proportion of actual positives correctly identified). Liblinear SVC

achieved a test accuracy of 72% and test precision of 58%. Remaining SVC variants along with both ensemble classifiers showed test accuracy ranging from 61–67% and test precision ranging from 50–54%. Decision Tree and Random Forest showed relatively lower test accuracy (56% and 50% respectively), with Random Forest exhibiting the characteristic pattern of high training accuracy (100%) alongside reduced test performance, indicative of overfitting in the absence of additional regularization. KNN achieved a test accuracy of 61% with a test recall of 71%, while Gaussian Naïve Bayes and Averaged Stochastic Gradient Descent each achieved 61% test accuracy with 86% test recall, demonstrating competitive sensitivity despite lower overall accuracy relative to the SVC-based models.

For the A1DNN, both atlases produced a test recall of 54%. CC200 outperformed EZ on test accuracy (63% vs. 53%) and test precision (61% vs. 49%). See Figures 1, 2, and 3 for full model results by atlas.

Figure 1 — Machine Learning Model Results: EZ Atlas

Method	Train Acc.	Train Prec.	Train Recall	Test Acc.	Test Prec.	Test Recall
KNN	72%	74%	74%	61%	50%	71%
Multiclass Logistic Regression	68%	69%	74%	67%	54%	100%
Decision Tree	74%	75%	76%	56%	45%	71%
Random Forest	100%	100%	100%	50%	43%	86%
Linear SVC	68%	69%	74%	67%	54%	100%
Liblinear SVC	68%	69%	74%	72%	58%	100%
Gaussian RBF Kernel SVC	72%	72%	78%	61%	50%	100%
Averaged Stochastic Gradient Descent	68%	68%	74%	72%	60%	86%
Gaussian Naive Bayes	69%	72%	69%	61%	50%	86%
Voting Classifier	70%	71%	75%	61%	50%	100%
Stacking Classifier	87%	87%	90%	61%	50%	100%

Figure 1: Machine Learning Model Results using the EZ Atlas.

Figure 2 — Machine Learning Model Results: CC200 Atlas

Method	Train Acc.	Train Prec.	Train Recall	Test Acc.	Test Prec.	Test Recall
KNN	72%	74%	74%	61%	50%	71%
Multiclass Logistic Regression	68%	69%	74%	67%	54%	100%
Decision Tree	74%	75%	76%	56%	45%	71%
Random Forest	100%	100%	100%	50%	43%	86%
Linear SVC	68%	69%	74%	67%	54%	100%
Liblinear SVC	68%	70%	74%	72%	58%	100%
Gaussian RBF Kernel SVC	72%	72%	78%	61%	50%	100%
Averaged Stochastic Gradient Descent	68%	68%	74%	72%	60%	86%
Gaussian Naive Bayes	69%	72%	69%	61%	50%	86%
Voting Classifier	70%	71%	75%	61%	50%	100%
Stacking Classifier	87%	87%	90%	61%	50%	100%

Figure 2: Machine Learning Model Results using the CC200 Atlas.

Figure 3 — A1DNN Performance Metrics: EZ vs. CC200 Atlas

Model	Method	Train Acc.	Sensitivity	Specificity	Test Acc.	Test Prec.	Test Recall
EZ Atlas DNN	A1DNN	63%	61%	61%	53%	49%	54%
CC200 Atlas DNN	A1DNN	66%	60%	68%	63%	61%	54%

Figure 3: Comparison of EZ and CC200 A1DNN Performance Metrics.

5. DISCUSSION

The present study investigated ASD classification from resting-state fMRI data using two brain atlases (EZ and CC200), a battery of machine learning classifiers, and a basic digital neural network (A1DNN). Results revealed that atlas selection had a differential effect depending on model type: machine learning model performance was largely equivalent across atlases, while the A1DNN demonstrated meaningfully better performance with the CC200 atlas relative to EZ.

Across all machine learning models and both atlases, Liblinear SVC achieved the highest test accuracy (72%), consistent with the broader literature establishing SVM variants as among the strongest performers for ASD classification from fMRI data [9, 10, 12]. The 100% test recall observed across Liblinear SVC, Linear SVC, and ensemble classifiers (Voting

and Stacking) is notable: in a clinical screening context, high recall — minimizing false negatives — is arguably more important than precision, since missing a true ASD case carries greater practical cost than a false positive that prompts further evaluation [2, 4]. The trade-off observed here, where precision ranged from 50–58% alongside perfect recall, reflects a known tension in imbalanced classification tasks and is consistent with findings in comparable studies [5, 12].

The equivalence of machine learning model performance across atlases aligns with prior work suggesting that atlas choice does not straightforwardly determine ML classifier outcome, as the feature extraction and dimensionality reduction pipeline — here, PCA followed by RFECV with AdaBoost — may normalize atlas-level differences prior to classification [6]. By contrast, the A1DNN showed sensitivity to atlas selection, with CC200 outperforming EZ on both accuracy (63% vs. 53%) and precision (61% vs. 49%) while recall remained equivalent (54%). This is consistent with prior findings highlighting CC200 and Craddock atlases as particularly well-suited for neural network approaches to ABIDE data [11], and may reflect the CC200 parcellation capturing connectivity patterns that are more informative for learned feature representations in dense network architectures.

The use of PCA followed by RFECV addresses a limitation noted across the literature: standard dimensionality reduction via PCA alone ignores class labels, potentially retaining variance that is uninformative for classification [5, 6]. The addition of RFECV with AdaBoost as the meta-estimator provides a supervised refinement step that iteratively selects features with discriminative value, a hybrid approach that extends prior SVM-RFE based pipelines [9, 10] while avoiding the computational constraints associated with SVM-based selection on high-dimensional data.

Several limitations of the present work should be noted. The ABIDE dataset carries well-documented biases toward male and higher-functioning individuals, which constrains the generalizability of classification performance to the broader ASD population [4, 8]. The A1DNN architecture employed here is intentionally basic — a sequential dense network

— and does not incorporate the spatial or temporal structure of fMRI data that CNN or RNN/LSTM architectures are designed to exploit [13, 14, 17]. Performance metrics for the A1DNN are accordingly modest, and this architecture is best understood as a baseline for structured comparison rather than a performance benchmark. Future work will extend atlas comparisons across the full set of ABIDE atlases using the same pipeline, providing a more systematic picture of whether the ML/DNN divergence in atlas sensitivity observed here generalizes across parcellation schemes.

From a translational perspective, the high recall achieved by Liblinear SVC and ensemble classifiers carries practical significance for clinical screening contexts. A classifier that minimizes false negatives — that is, rarely misses a true ASD case — supports a triage model in which positive classifications prompt further structured evaluation rather than serving as a definitive diagnosis. The precision-recall trade-off observed here, where high recall was accompanied by moderate precision, is consistent with the design priorities appropriate for a screening instrument rather than a diagnostic one [2]. Continued development of rs-fMRI-based classification pipelines should therefore weigh sensitivity to missed cases as a primary optimization target, alongside efforts to improve dataset representativeness across sex, functioning level, and cultural background.

6. CONCLUSION

This study applied a PCA-RFECV feature selection pipeline followed by a battery of machine learning classifiers and an A1DNN to resting-state fMRI data from the EZ and CC200 atlases for ASD prediction. The central finding is that atlas selection produced differential effects depending on model type: machine learning classifier performance was equivalent across both atlases, while the A1DNN showed meaningfully better performance with CC200 than EZ. Liblinear SVC was the best-performing machine learning classifier across both atlases, and CC200 was the better-performing atlas for the A1DNN. These results suggest that the PCA-RFECV pipeline effectively normalizes atlas-level differences for classification-

based models, while neural network architectures retain sensitivity to atlas parcellation. The practical significance of this work lies in demonstrating that computational budget optimization enables broader cross-comparative evaluation without sacrificing model coverage, and that atlas choice is a meaningful variable for neural network approaches. Future research will extend these comparisons across all ABIDE atlases using the same pipeline to determine whether the pattern of ML stability and DNN atlas sensitivity holds more broadly across different parcellation schemes.

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Data Availability and Ethics Statement: This study used the Autism Brain Imaging Data Exchange (ABIDE) dataset, a publicly available, de-identified repository. No new data were collected. No institutional review board approval was required. The ABIDE dataset is available at `http://fcon_1000.projects.nitrc.org/indi/abide/`.

Ethics Approval and Consent to Participate: This study used only publicly available, fully de-identified data from the ABIDE repository. No human participants were recruited, and no new data were collected. Formal ethics approval was therefore not required.

Human and Animal Rights: No animals were used in this study. All procedures involving human data conformed to the ethical standards of the institutions that originally collected the ABIDE data, in accordance with the Declaration of Helsinki.

Consent for Publication: Not applicable. No individually identifiable data are reported.

Standards of Reporting: This study follows the TRIPOD (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis) reporting guidelines for prediction model development studies.

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